English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 21.01.2024 Test 2

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

m, n, k denote nonnegative integers.

Problem 1. Let X have distribution function

$$F_X(x) = \begin{cases} 0, & \text{for } x \leq 0; \\ ax^2 + bx, & \text{for } 0 < x \leq 1; \\ 1, & \text{for } x > 1. \end{cases}$$

1. (0.25 points) Is X a continuous random variable? If so, derive its density.
2. (0.25 points) Find a, b if additionally $\mathbb{P}(X > 0.75) = 0.5$.
3. (0.25 points) If the expectation of a random variable Y exists, then

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y > t)\mathbb{D}t - \int_{-\infty}^0 \mathbb{P}(Y \leq t)\mathbb{D}t.$$

   Prove this identity when Y is continuous with density f_Y.

4. (0.25 points) Use the formula to find $\mathbb{E}X$.

Solution 1.

1. Yes: it is continuous because it takes uncountably many values. It is also absolutely continuous because the given distribution function is differentiable:

$$f_X(x) = \frac{d}{dx}F_X(x) = \begin{cases} 0, & \text{for } x \leq 0; \\ 2ax + b, & \text{for } 0 < x < 1; \\ 0, & \text{for } x > 1. \end{cases} = (2ax + b)\mathbb{1}_{\{0<x<1\}}$$

2. We are also given $\mathbb{P}(X > 0.75) = 1 - F_X(0.75) = 0.5$. From the definition, $1 = \mathbb{P}(X < 1) = F_X(1) = a + b$. Solve the system:

$$\left| \begin{array}{l} 1 - F_X\left(\frac{3}{4}\right) = \frac{1}{2} \\ F_X(1) = 1 \end{array} \right. \Leftrightarrow \left| \begin{array}{l} F_X\left(\frac{3}{4}\right) = \frac{1}{2} \\ F_X(1) = 1 \end{array} \right. \Leftrightarrow \left| \begin{array}{l} \frac{9}{16}a + \frac{3}{4}b = \frac{1}{2} \\ a + b = 1 \end{array} \right.$$

   Thus $(a, b) = \left(\frac{1}{6}, \frac{5}{6}\right)$.

3. The alternative expression for the expectation follows by interchanging the order of integration. We show that the right side equals the left. Consider the first summand:

$$\int_0^\infty \mathbb{P}(Y > t)dt = \int_0^\infty \left(\int_t^\infty f_Y(y)dy\right) dt = \int\limits_{\{0<y<t\}} f_Y(y)d(t, y) = \int_0^\infty \left(\int_0^y f_Y(y)dt\right) dy =$$

$$\int_0^\infty \left(\int_0^y dt\right) f_Y(y)dy = \int_0^\infty y f_Y(y)dy$$

   Similarly, for the second summand,

$$-\int_{-\infty}^0 \mathbb{P}(Y \leq t)dt = \int_{-\infty}^0 y f_Y(y)dy$$

   completing the proof.

   Formally, we used Fubini's theorem, which allows the order of integration to be interchanged when the integral over the region is absolutely convergent. Explain why this applies here.

4.

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t)dt - \int_{-\infty}^0 \mathbb{P}(X < t)dt$$

$$= \int_0^\infty (1 - F_X(t))dt - \underbrace{\int_{-\infty}^0 F_X(t)dt}_{=0}$$
$$= \int_0^1 (1 - ax^2 - bx)dx + \underbrace{\int_1^\infty (1 - F_X(x))dx}_{=0}$$
$$= 1 - \frac{a}{3} - \frac{b}{2}$$
$$= \frac{19}{36}$$

To check the arithmetic, compare the result with the usual calculation $\int_0^1 x f_X(x)\,dx$.

Problem 2. Let $U_1, U_2, U_3 \sim U(0,1)$ be independent, and let $U_{(1)}, U_{(2)}, U_{(3)}$ be the values U_1, U_2 and U_3 in increasing order.

For example, if $U_1 = 0.23$, $U_2 = 0.88$, $U_3 = 0.1$, *then* $U_{(1)} = 0.1$, $U_{(2)} = 0.23$, $U_{(3)} = 0.88$.

1. (0.25 points) Find the densities and expectations of $U_{(1)} := \min\{U_1, U_2, U_3\}$ and $U_{(3)} := \max\{U_1, U_2, U_3\}$.
2. (0.25 points) Is it true that $U_{(2)} \sim U(0,1)$? If yes, prove it; otherwise find its distribution.
3. (0.25 points) Find the variance of $X := -\ln(U_1)$.

Let $X_1, X_2, X_3, X_4 \sim Exp(1)$ be independent.

4. (0.25 points) Find the densities, expectations and variances of $S_2 := X_1 + X_2$, $S_3 := X_1 + X_2 + X_3$ and $S_4 := X_1 + \cdots + X_4$.
5. (0.5 points) Let $Y_k := S_k/S_4$. Prove that $Y_{(k)}$ and $U_{(k)}$ have the same distribution for $k = 1, 2, 3$.

Solution 2.

1. Proceed in the standard way (for $0 < t < 1$; outside this interval the result is immediate):

$$F_{U_{(1)}}(t) = \mathbb{P}(U_{(1)} < t) = 1 - \mathbb{P}(U_{(1)} > t) = 1 - \mathbb{P}(U_1 > t)^3 = 1 - (1-t)^3$$
$$\Rightarrow f_{U_{(1)}}(t) = \frac{d}{dt} F_{U_{(1)}}(t) = -3(1-t)^2 \cdot (-1) = 3(1-t)^2$$
$$\Rightarrow E[U_{(1)}] = \int_0^1 t \cdot 3(1-t)^2 dt = 3\int_0^1 (t^3 - 2t^2 + t)dt = \cdots = \frac{1}{4}$$

$$F_{U_{(3)}}(t) = \mathbb{P}(U_{(3)} < t) = \mathbb{P}(U_1 < t)^3 = t^3$$
$$\Rightarrow f_{U_{(3)}}(t) = 3t^2$$
$$\Rightarrow E[U_{(3)}] = \int_0^1 t \cdot 3t^2 dt = 3\int_0^1 t^3 dt = \cdots = \frac{3}{4}$$

2. It is worth considering why the initial guess is misleading, but directly:

$$F_{U_{(2)}}(t) = \mathbb{P}(U_{(2)} < t)$$
$$= \mathbb{P}\left(\text{at least two of the three independent values are } < t\right)$$
$$= \mathbb{P}(Bin(3,t) \geq 2)$$
$$= \binom{3}{2} t^2(1-t) + t^3$$
$$= 3t^2 - 2t^3$$
$$\Rightarrow f_{U_{(2)}}(t) = 6t - 6t^2 = 6t(1-t)$$

3. Write $U := U_1$. We have $X := -\ln(U) =: g(U) \in [0, +\infty)$. The function g is strictly decreasing and hence invertible; its inverse $h(x) := g^{-1}(x) = e^{-x}$ is differentiable. Changing variables,

$$f_X(x) = \underbrace{f_U(h(x))}_{=1} \cdot |\underbrace{h'(x)}_{-e^{-x}}| = e^{-x} \text{ for } 0 < x.$$

Hence $X \sim Exp(1)$, and we may use $\mathbb{E}[X] = DX = 1$ or calculate directly:

$$\begin{aligned}\mathbb{E}[X] &= \int_0^\infty x e^{-x} dx = \cdots = 1\\ \mathbb{E}[X^2] &= \int_0^\infty x^2 e^{-x} dx = \cdots = 2\\ \Rightarrow \mathbb{D}[X] &= 2 - 1 = 1\end{aligned}$$

4. Recall that a sum of independent identically distributed exponential random variables is gamma-distributed. If $X_i \sim Exp(\lambda)$ are independent, then

$$S_n := \sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$$

Thus, in general and (on the right) in our particular case, for $t > 0$:

$$\begin{aligned} f_{S_n}(t) &= \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} && = \frac{t^{n-1}}{(n-1)!} e^{-t}\\ \mathbb{E}[S_n] &= \frac{n}{\lambda} && = n\\ \mathbb{D}[S_n] &= \frac{n}{\lambda^2} && = n\end{aligned}$$

The expectation and variance follow directly from independence. Even without using the gamma distribution, the density can be derived from the convolution formula for a sum of two independent random variables:

$$f_{S_2}(t) \overset{t\geq 0}{=} \int_{-\infty}^{+\infty} \underbrace{f_{X_1}(x)}_{=e^{-x}\cdot \mathbb{1}_{\{0<x\}}} \underbrace{f_{X_2}(t-x)}_{e^{-(t-x)}\cdot \mathbb{1}_{\{0<t-x\}}} dx = \int_0^t e^{-x} e^{x-t} dx = e^{-t} \int_0^t dx \qquad = te^{-t}$$

$$f_{S_3}(t) \overset{t\geq 0}{=} \int_{-\infty}^{+\infty} \underbrace{f_{S_2}(x)}_{=xe^{-x}\cdot \mathbb{1}_{\{0<x\}}} \underbrace{f_{X_3}(t-x)}_{e^{-(t-x)}\cdot \mathbb{1}_{\{0<t-x\}}} dx = \int_0^t x e^{-x} e^{x-t} dx = e^{-t} \int_0^t x dx \qquad = \frac{t^2}{2} e^{-t}$$

$f_{S_4}(t) \overset{t\geq 0}{=} \ldots$ by induction.

5. For the last part, write

$$Y_k = \frac{X_1 + \cdots + X_k}{(X_1 + \cdots + X_k) + (X_{k+1} + \cdots + X_4)} = \frac{S_k}{S_k + \widetilde{S}_{4-k}},$$

where $\widetilde{S}_{4-k}$ is a copy of S_{4-k} independent of S_k. Then use the standard method for the density of a transformation (a two-dimensional change of variables, its Jacobian, etc.).

Problem 3. 1. (0.25 points) In the manufacture of specialized automotive parts, the lengths of certain components are measured, and deviations from the recommended length are modelled with mean 0 and variance 9. A component is acceptable if its deviation from the recommended length is less than a threshold s. What percentage of length errors fall within the acceptable range when $s = 1$? Find s such that 95% of components meet the acceptance criterion.

2. (0.5 points) Let $X \sim N(1, 1)$. To accuracy 0.01, find $\mathbb{E}|X|$.

Solution 3. 1.

2.

$$\begin{aligned}\mathbb{E}[|X|] &= \int_{-\infty}^{+\infty} |x| f_X(x) dx \overset{t:=x-1}{=} \int_{-\infty}^{+\infty} |t+1| \underbrace{f_X(t+1)}_{f_Z(t)} dt\\ &= \int_{-\infty}^{-1} -(t+1) f_Z(t) dt + \int_{-1}^{+\infty} (t+1) f_Z(t) dt\\ &= \underbrace{-\int_{-\infty}^{-1} f_Z(t) dt}_{=-\Phi(-1)} + \underbrace{\int_{-1}^{+\infty} f_Z(t) dt}_{=1-\Phi(-1)} - \underbrace{\int_{-\infty}^{-1} t f_Z(t) dt}_{t\mapsto -t} + \underbrace{\int_{-1}^{+\infty} t f_Z(t) dt}_{=\int_{-1}^{-1} + \int_1^\infty}\end{aligned}$$

$$= 1 - 2\underbrace{\Phi(-1)}_{1-\Phi(1)} + 2\int_1^\infty t f_Z(t)\,dt + \underbrace{\int_{-1}^1 t f_Z(t)\,dt}_{=0 \text{ (odd)}}$$
$$= 2\Phi(1) - 1 + 2J$$

It remains to calculate:

$$J := \int_1^\infty t f_Z(t)dt = \int_1^\infty t \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = -\frac{1}{\sqrt{2\pi}} \int_1^\infty e^{-\frac{t^2}{2}} d\left(-\frac{t^2}{2}\right) = -\frac{1}{\sqrt{2\pi}} \left[e^{-\frac{t^2}{2}}\right]_{t=1}^{t\to+\infty} = \frac{1}{\sqrt{2\pi e}}$$

Finally:

$$\mathbb{E}[|X|] = -1 + 2\underbrace{\Phi(1)}_{\approx 0.841} + \underbrace{\frac{2}{\sqrt{2\pi e}}}_{\approx 0.484} \approx 1.166$$

Problem 4. The plane is ruled by parallel lines spaced $2t$ apart. A needle of length $2l \le 2t$ is dropped at a random position.

1. (0.5 points) What is the probability that the needle touches a line?

   Hint: Parametrize by the position of the needle's centre and its angle to the lines.

2. (0.25 points) Suppose $l = t$. What minimum number of independently dropped needles makes the probability that at least half cross a line at least 95%? Explain how and why this procedure can estimate π.

3. *(0.25 points) Bonus: Solve part 1 when $l > t$.*

Solution 4. This is known as Buffon's needle problem.

1. Let

   $$\begin{aligned} X &:= \text{distance from the needle's centre to the nearest line} && \sim U(0,t) \\ \theta &:= \text{acute angle between the needle and the nearest line} && \sim U\left(0, \frac{\pi}{2}\right). \end{aligned}$$

   The needle crosses the line if and only if $X \le l\sin\theta$, so we seek:

   $$p := \mathbb{P}(X \le l\sin\theta) = \int_0^{\frac{\pi}{2}} \int_0^{l\sin\theta} \frac{2}{\pi t} dt d\theta = \frac{2l}{\pi t}$$

   See the linked solution of Buffon's needle problem.

2. Since $l = t$, $p = 2/\pi$. Model the needle drops as Bernoulli trials: $S_n := \sum_{i=1}^n X_i$, where the independent $X_i \sim Ber(p)$ indicate success when a needle crosses a line. We seek an estimate of n such that

   $$\mathbb{P}\left(S_n \ge \frac{1}{2}n\right) \ge 95\%$$

   Use the CLT. First note $\mu := \mathbb{E}[X_1] = p = 2/\pi$ and $\sigma := \sqrt{\mathbb{D}[X]} = \sqrt{p(1-p)}$. We require:

   $$\mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \ge \frac{\frac{n}{2} - n\mu}{\sigma\sqrt{n}}\right) = \mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \ge \frac{\sqrt{n}(\frac{1}{2} - \mu)}{\sigma}\right) \overset{\text{CLT}}{\approx} \mathbb{P}\left(Z \ge \underbrace{\frac{\sqrt{n}\left(\frac{1}{2} - \mu\right)}{\sigma}}_{=:C_n}\right) \ge 95\%$$

   $$\begin{aligned} 1 - \Phi(C_n) &\ge 95\% \\ \Phi(C_n) &\le 5\% && \Phi \text{ is increasing, as is } \Phi^{-1} \\ C_n &\le \Phi^{-1}(5\%) \\ \frac{\sqrt{n}(\frac{1}{2} - \mu)}{\sigma} &\le \Phi^{-1}(5\%) && C_n < 0 \text{ and decreases with } n, \text{ since } \tfrac{1}{2} < \mu \\ \sqrt{n} &\ge \frac{\sigma\Phi^{-1}(5\%)}{\frac{1}{2} - \mu} && \text{note also } \Phi^{-1}(5\%) < 0 \end{aligned}$$

$$n \geq \left(\frac{\sigma \Phi^{-1}(5\%)}{\frac{1}{2} - \mu} \right)^2$$

Approximating:

$$\begin{aligned} \Phi^{-1}(5\%) &\approx -1.645 \\ \sigma &\approx 0.481 \\ \mu &\approx 0.637 \\ \Rightarrow \left(\frac{\sigma \Phi^{-1}(5\%)}{\frac{1}{2} - \mu} \right)^2 &\approx 33.5387 \end{aligned}$$

Finally, $n \geq 34$.

The (strong) law of large numbers explains the estimate of π. Since $\frac{1}{n} S_n \xrightarrow{\text{a.s.}} \mu = 2/\pi$, estimate π by $2/(\frac{1}{n} S_n)$. Consider how large n should be for a desired accuracy.

3. See the long-needle case in the linked solution of Buffon's needle problem.